

Lab Exercise 19

TRIGGER

AIM

To implement a **BEFORE UPDATE trigger** in MySQL to record the old values of a vegetable before they are changed.

PROCEDURE

1. Select the `VegetableDB` database.
2. Create the `Vegetable` table.
3. Create a `Vegetable_Audit` table to store changes.
4. Create a **BEFORE UPDATE trigger** on the `Vegetable` table.
5. Update a vegetable record.
6. The trigger automatically stores the old values in the audit table.
7. Display the audit table.

1. Select Database

```
USE VegetableDB;
```

2. Create Vegetable Table

```
CREATE TABLE Vegetable (  
    VegID INT PRIMARY KEY,  
    VegName VARCHAR(30),  
    Category VARCHAR(30),  
    Price DECIMAL(10,2),  
    Quantity INT  
);
```

3. Insert Records

```
INSERT INTO Vegetable VALUES  
(1, 'Tomato', 'Vegetable', 30.50, 100),  
(2, 'Potato', 'Vegetable', 25.00, 200),  
(3, 'Carrot', 'Root', 40.00, 80),  
(4, 'Onion', 'Vegetable', 35.00, 150),  
(5, 'Brinjal', 'Vegetable', 45.50, 90),  
(6, 'Cabbage', 'Leafy', 50.00, 70);
```

4. Create Audit Table

```
CREATE TABLE Vegetable_Audit (  
    VegID INT,  
    OldVegName VARCHAR(30),  
    OldCategory VARCHAR(30),  
    OldPrice DECIMAL(10,2),  
    OldQuantity INT  
);
```

5. Create BEFORE UPDATE Trigger

```
DELIMITER //
```

```
CREATE TRIGGER before_vegetable_update
BEFORE UPDATE ON Vegetable
FOR EACH ROW
BEGIN
    INSERT INTO Vegetable_Audit
    VALUES (
        OLD.VegID,
        OLD.VegName,
        OLD.Category,
        OLD.Price,
        OLD.Quantity
    );
END //
```

```
DELIMITER ;
```

6. Update a Vegetable

```
UPDATE Vegetable
SET Price = 55.00,
    Quantity = 120
WHERE VegID = 1;
```

The trigger automatically stores the **old values** of Tomato.

7. Display Updated Vegetable

```
SELECT * FROM Vegetable
WHERE VegID = 1;
```

```
mysql> SELECT * FROM Vegetable
-> WHERE VegID = 1;
+-----+-----+-----+-----+-----+
| VegID | VegName | Category | Price | Quantity |
+-----+-----+-----+-----+-----+
|      1 | Tomato  | Vegetable | 55.00 |      120 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

8. Display Audit Table

```
SELECT * FROM Vegetable_Audit;
```

```
mysql> SELECT * FROM Vegetable_Audit;
+-----+-----+-----+-----+-----+
| VegID | OldVegName | OldCategory | OldPrice | OldQuantity |
+-----+-----+-----+-----+-----+
|      1 | Tomato     | Vegetable   |    30.50 |          100 |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

